

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Inclusive Rice Transplanting Mechanization Services in Bangladesh

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.2 · Entrepreneurship through Mechanization & Postharvest Services

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

In Bangladesh, rice is the cornerstone of food security, cultivated on over 11 million hectares annually. However, the sector faces a critical labor crisis as rural workers migrate to cities, leaving an aging workforce unable to perform timely transplanting. This leads to increased costs, reduced yields, and declining profitability for small-holder farmers. This solution track aims to scale an inclusive, market-led machinery service provision (MSP) model to deliver affordable and timely rice transplanting services, addressing a national bottleneck and building a sustainable mechanization ecosystem.

The core innovation is a bundled package centered on **Rice Transplanter-based Machinery Service Provision (RT-MSP)**. Instead of individual farmers owning expensive machines, trained local entrepreneurs (MSPs) purchase rice transplanters and offer transplanting services for a fee. This model is uniquely complemented by **women-led seedling enterprises** that use tray-based seed sowing machines to ensure a reliable supply of the high-quality seedlings required for mechanical transplanting. This holistic approach is anchored by strong private-sector supply chains for machinery, spare parts, and after-sales service, and facilitated by blended finance solutions.

This scaling track builds on over a decade of successful implementation and evidence generation by CIMMYT through bilateral projects like CSISA-MEA. The model is mature, with proven adoption drivers and clear profitability for all actors. With catalytic support from S4I, this track will leverage strong partnerships with the Ministry of Agriculture (MoA), the Department of Agricultural Extension (DAE), leading agribusinesses (e.g., ACI Motors, The Metal, Abedin Equipment), microfinance institutions (MFIs), and a network of 700 local engineering workshops to transition from a subsidy-driven system to a self-sustaining, market-led model at national scale.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

- **Machinery Service Provision (MSP) Model:** An entrepreneurial model where private service providers own and operate rice transplanters, selling transplanting services to smallholder farmers.
- **Women-led Seedling Enterprises:** A parallel business model where women entrepreneurs use seed sowing machines to produce the tray-based seedlings essential for mechanical transplanters, creating a new value chain and income stream.
- **Private Sector Supply Chains:** Strengthened networks of machinery importers, district-level dealers, and a vast network of local light engineering workshops (ABLEs) for the production of spare parts and provision of after-sales services.
- **Blended Finance Solutions:** Facilitating access to credit and leasing products from MFIs and innovative finance organizations to overcome high upfront investment costs for MSPs and seedling entrepreneurs.

2.2 Evidence of Impact

- **Cost Savings:** Farmers save **25-30% (USD 55-60/ha)** in transplanting costs compared to manual labor.
- **Yield Increase:** Timely and uniform planting contributes to yield gains of up to 10%.
- **High ROI:** MSPs achieve a **35-40% return on investment**, with net annual profits of **USD 1,500-2,000** per machine.
- **Proven Adoption:** Demonstrations and pilots under the CSISA-MEA project established clear farmer willingness to pay for RT services when accessible.

- **Private Sector Demand:** Leading agribusiness firms have expressed strong demand for partnerships to expand the market.

2.3 Scaling Targets

Farm Reached	Households	30,000	150,000
Area under Transplanting	Mechanized	10,000 - 11,000 ha	50,000 - 55,000 ha
Active Rice MSPs	Transplanter	200	1,000
Women-led Enterprises	Seedling	100	500
Jobs (operators, mechanics)	Supported	~600	~3,000
Investment (private and public)	Leveraged	USD 1M+	USD 5M+

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy The scaling pathway is a **public-private partnership (PPP) model** designed to create a self-sustaining market system. It moves away from unsustainable direct subsidies and focuses on facilitating private-sector leadership. The key elements of the pathway are:

- **Demand Creation:** Multi-channel awareness campaigns (digital platforms, field demonstrations, extension services) will educate farmers on the economic and agronomic benefits of mechanical transplanting, shifting traditional mindsets.
- **Market Strengthening:** S4I will facilitate connections between MSPs, machinery companies, local fabricators (ABLEs), and financial institutions. This includes strengthening dealer networks for after-sales service and supporting local spare parts manufacturing to reduce costs and downtime.
- **Inclusive Entrepreneurship:** Targeted support, including training and access to tailored finance, will enable women to lead the new seedling enterprise value chain and encourage youth participation as machine operators.
- **Policy Engagement:** Evidence from the scaling process will be used to engage with the Ministry of Agriculture and DAE to advocate for policies that support service-based mechanization (e.g., performance-based incentives) over direct machine subsidies.

3.2 Theory of Change If smallholder farmers gain access to affordable, timely rice transplanting services provided by local entrepreneurs, **and** the supply of quality seedlings, machinery, spare parts, after-sales service, and finance is simultaneously strengthened through private-sector partnerships, **then** farmer adoption of mechanized transplanting will increase, leading to reduced production costs and higher incomes, **ultimately contributing** to a sustainable, inclusive market system for agricultural mechanization in Bangladesh that enhances food security and climate resilience.

3.3 Key Activities Funded by S4I

- **Awareness and Demand Creation:** Scaling farmer field demonstrations, deploying digital campaigns, and leveraging government extension networks to drive farmer demand for RT services.
- **Capacity Building:** Training programs for new MSPs (operators) and women seedling entrepreneurs on business management, machine operation, and quality seedling production.
- **Market Facilitation:** Strengthening linkages between MSPs, machinery dealers, and local fabricators to ensure robust after-sales service and spare parts supply chains.
- **Finance Facilitation:** Working with partner MFIs and finance institutions to develop and roll out tailored credit and leasing products for MSPs and seedling enterprises.
- **Evidence and Learning:** Implementing a MELIA framework to track adoption, profitability, and inclusion, and using learnings for adaptive management and policy engagement.

4. Partners and Roles

Demand Partners	Farmers: 150,000 farm households by 2028. MSPs: 1,000 machinery service providers. Seedling Entrepreneurs: 500 women-led enterprises.	Primary end-users and adopters of the innovation bundle. Their adoption and payment for services drive the entire market system.
Scaling Partners	Private Sector: 3-5 lead machinery firms (ACI Motors, The Metal, Abedin Equipment), 10-15 district dealers, 5-10 local manufacturers (SMR Agro, Al Madina Engineering). Finance Institutions: 5-10 partners (GUK, SDC, VPKA Foundation, LightCastle Partners). Public Sector: Ministry of Agriculture (MoA), Department of Agricultural Extension (DAE), Bangladesh Rice Research Institute (BRRI).	Private Sector: Supply machinery, provide after-sales service, market the technology, and provide in-kind support. Finance: Provide credit and leasing products to end-users. Public Sector: Provide technical support, training platforms, farmer outreach, and enabling policy frameworks.
CGIAR Champions	CIMMYT: Lead technical and delivery partner (Owen Calvert, Md. Kafil Uddin, Bharathi Parupalli). IRRI: Potential technical advice on seedling systems.	Provide technical leadership, facilitate partnerships, ensure responsible scaling, coordinate activities, and generate evidence for learning and policy influence.

Funding Model: S4I provides catalytic funding for **facilitation, coordination, awareness creation, and capacity building**. Private sector partners contribute through **in-kind support** (e.g., for joint awareness campaigns), and end-users finance the core inputs (machinery, seedlings) via their own resources or **third-party loans/leasing** facilitated by the project.

5. Enabling Environment and Risks

5.2 Opportunities

- **National Priority:** The Government of Bangladesh has a stated goal of expanding agricultural mechanization, creating a favorable policy environment.
- **Private Sector Appetite:** Leading agribusiness firms are eager to expand into the rice transplanter market and have provided letters of intent to collaborate.
- **Established Infrastructure:** A network of 700 trained local engineering workshops (ABLEs) is ready to produce spare parts, mitigating a key risk of machine downtime.
- **Shifting Subsidy Landscape:** The planned phase-out of direct machinery subsidies creates an opening for a more sustainable, market-led service model.

5.2 Challenges and Mitigation

- **Financial Risk:** Farmers and entrepreneurs may struggle with upfront costs.
- **Mitigation:** Facilitate blended finance (credit, leasing) with MFIs; promote the proven 35-40% ROI to build investor confidence.
- **Adoption Bottleneck:** Farmers are traditionalists and may be skeptical of new technology and the temporary "shock" to seedlings post-transplanting.
- **Mitigation:** Intensive, multi-channel awareness campaigns, including live field demonstrations at harvest to visually prove yield benefits.
- **Policy Uncertainty:** Changes in import taxes or subsidy structures could destabilize the market.
- **Mitigation:** Proactive policy engagement using evidence from the field to advocate for supportive, stable policies.
- **Social/Gender Barriers:** Women may face challenges entering the mechanization value chain.
- **Mitigation:** Deliberate focus on women-led seedling enterprises, gender training, and developing finance products accessible to women.
- **Operational Risk:** The ambitious 2026 target of financing 200 RTs may be hampered by cautious lenders.
- **Mitigation:** Early and intensive engagement with financial partners to de-risk the investment and build confidence in the business model.

6. Learning and Adaptation

6.1 Monitoring Focus

- **Process Outcomes:** Effectiveness of awareness campaigns; number of partnerships formed with private sector and finance institutions.
- **Behavior Change:** Shifts in farmer willingness to pay for RT services; increased confidence and investment by machinery dealers and MFIs in the RT market.
- **System Outcomes:** Increase in number of active MSPs and seedling enterprises; hectares covered; jobs created (disaggregated by women/youth); volume of private sector co-investment and credit facilitated.

6.2 Adaptive Management

- **Quarterly Review Meetings:** Formal check-ins with the S4I team to review progress against milestones, discuss challenges, and adapt strategies based on real-time evidence.
- **Annual PORB Cycle:** Evidence from the annual MELIA survey will feed into the program's annual Planning, Outcome, Reporting, and Budgeting cycle to inform planning for the subsequent year.
- **Peer Learning:** Active engagement with other S4I solution tracks (e.g., digital advisory) to explore synergies and share best practices in scaling.

7. Next Steps and Timeline (2026)

Q1 (Jan-Mar)	Finalize partner MOUs with private sector and finance institutions; launch baseline assessment and refine MELIA tools; begin selection process for first cohort of MSPs and seedling entrepreneurs.
Q2 (Apr-Jun)	Roll out "Boro" season awareness campaigns and field demonstrations; conduct first training of trainers for new MSPs; facilitate finance linkages for first 200 RTs.
Q3 (Jul-Sep)	Support "Aman" season service provision; monitor adoption and gather feedback; conduct mid-term review and adaptive management workshop; initiate policy engagement with DAE.
Q4 (Oct-Dec)	Conduct end-line annual survey to measure progress against 2026 targets; document success stories and lessons learned; develop detailed work plan and targets for 2027.

Communications and Knowledge: Success stories of lead MSPs (e.g., the entrepreneur who scaled from 1 to 9 machines) and women seedling entrepreneurs will be documented. Results will be shared via the CGIAR Results Dashboard, blog posts, and presentations to key stakeholders like the Ministry of Agriculture and donor partners.

8. Contact

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